SOLUTION

Midterm Exam II (120 minute exam)

Course: Electromagnetic Theory **Date:** April 22
Instructor: Dr. Shafayat Abrar **Marks:** 100 Marks

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Instructions:

1. Attempt all questions. Formula-sheet is **NOT** allowed.
2. Remember that talking is not allowed at any time in the exam hall.
3. Keep your eyes on your own paper. Remember, copying is cheating!
4. You are not permitted to share calculators or any other material during the examination.
5. Stop writing when the teacher says it is the end of the exam.

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Formulae:

- $E_{1t} = E_{2t} \quad \Rightarrow \quad \dfrac{D_{1t}}{\varepsilon_1} = \dfrac{D_{2t}}{\varepsilon_2}$
- $D_{1n} - D_{2n} = \rho_s \quad$ or $\quad D_{1n} = D_{2n} \;$ (if $\rho_s = 0$)
- $\dfrac{\tan\theta_1}{\tan\theta_2} = \dfrac{\varepsilon_{r1}}{\varepsilon_{r2}}$
- $\nabla \cdot \mathbf{J} = -\dfrac{\partial \rho_v}{\partial t}, \quad \rho_v(t) = \rho_v(0)\exp(-t/T_r), \quad T_r = \varepsilon/\sigma.$
- $\nabla^2 V = -\dfrac{\rho_v}{\varepsilon}$
- $\mathbf{E} = -\nabla V, \quad \mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}, \quad \mathbf{P} = \chi_e\varepsilon_0\mathbf{E}, \quad \varepsilon_r = 1 + \chi_e = \dfrac{\varepsilon}{\varepsilon_0}$
- $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv = Q_{\text{enc}} = \psi$
- $\oint_L \mathbf{E} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}$
- $\rho_v = \nabla \cdot \mathbf{D}, \qquad \rho_s = \mathbf{a}_n \cdot \mathbf{D}$
- $\rho_{pv} = -\nabla \cdot \mathbf{P}, \qquad \rho_{ps} = \mathbf{a}_n \cdot \mathbf{P}$
- $R = \dfrac{\ell}{\sigma S}, \quad$ or $\quad R = \dfrac{V}{I} = \dfrac{\int \mathbf{E} \cdot d\mathbf{l}}{\int \sigma\mathbf{E} \cdot d\mathbf{S}}$
- $\mathbf{J} = \sigma\mathbf{E}, \quad \mathbf{J} = \rho_v\mathbf{u}, \quad I = \int_S \mathbf{J} \cdot d\mathbf{S}$
- $P = \int (\mathbf{E} \cdot \mathbf{J})\, dv = VI = I^2R = V^2/R. \quad W_E = \frac{1}{2}\int_v (\mathbf{D} \cdot \mathbf{E})\, dv$
- $\varepsilon_0 = \dfrac{10^{-9}}{36\pi}$ [F/m]. $\; k = \dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^{-9}$ [m/F]

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Spring 2017.

1. (10 points) A lead ($\sigma = 5 \times 10^6$ S/m) bar of circular cross section has a hole bored along its length of 4m so that its cross section becomes that of Fig. 1, where the inner and outer radii are 1cm and 3cm, respectively. Find the resistance between the two ends of the (composite) bar if the hole is completely filled with copper ($\sigma = 5.8 \times 10^6$ S/m).

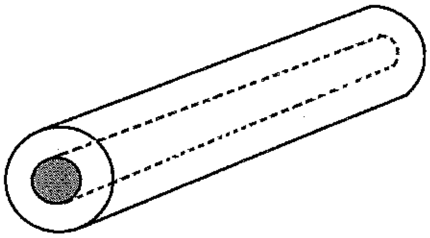

Fig. 1

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Let the resistance of lead be R_l

Let the resistance of copper be R_c

The net/total resistance is $R_l \| R_c = \dfrac{R_l R_c}{R_l + R_c}$

Resisance of Cu $= R_c = \dfrac{l}{\sigma_c S_c} = \dfrac{4}{5.8 \times 10^6 \, \pi \left(\frac{1}{100}\right)^2}$

$$= \frac{4 \times 10^4}{5.8 \times 10^6 \times \pi}$$

$$= \frac{4}{580\pi} \qquad = 2.1952 \, m\Omega$$

Resistance of Lead $= R_l = \dfrac{l}{\sigma_c (S_T - S_c)}$

$$= \frac{4}{5 \times 10^6 \, \pi \left(\left(\frac{3}{100}\right)^2 - \left(\frac{1}{100}\right)^2\right)}$$

$$= \frac{4 \times 10^4}{8 \times 5 \times 10^6 \, \pi} = \frac{1}{\pi} m\Omega = 0.3183 m\Omega$$

$R_T = R_l R_c / (R_l + R_c) = 278 \mu\Omega$.

2. (10 points) For the current density $\mathbf{J} = 10\rho\, z\, \mathbf{a}_\rho\, \mathrm{A/m^2}$. Find current through the cylindrical surface $\rho = 2$, $0 \le z \le 4\mathrm{m}$.

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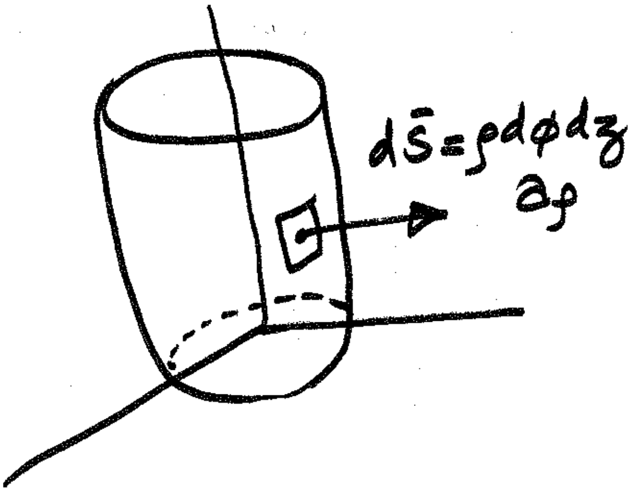
$$\bar{J} = 10\rho z\, a_\rho\, A/m^2$$

$$I = \int \bar{J} \cdot d\bar{S} = \int 10\rho^2 z\, d\phi\, dz$$



$$\int_{z=0}^{4}\int_{\phi=0}^{2\pi} 10\rho^2 z\, dz\, d\phi$$

$$= 10\rho^2 \cdot \frac{z^2}{2}\Big|_0^4 \cdot \phi\Big|_0^{2\pi} \quad (\rho = 2)$$

$$= \frac{40}{2}(16)(2\pi) = 16 \times 40\pi = \underline{640\pi\ A}$$

3. (10 points) In a dielectric material, $E_x + E_y + E_z = 10\text{V/m}$, and $\mathbf{P} = \frac{1}{30}(3\mathbf{a}_x + 2\mathbf{a}_y + 5\mathbf{a}_z)\ \text{nC/m}^2$. Calculate (a) χ_e, (b) $\mathbf{E}$, and (c) $\mathbf{D}$.

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(a) $\bar{P} = \chi_e \varepsilon_0 \bar{E} = \chi_e \varepsilon_0 (E_x a_x + E_y a_y + E_z a_z)$

$$= \frac{1}{30}(3a_x + 2a_y + 5a_z)$$

Compairing the above two expressions, we obtain

$$\chi_e \varepsilon_0 E_x = \frac{1}{10} \Rightarrow E_x = \frac{1}{10\chi_e\varepsilon_0}$$

$$\chi_e \varepsilon_0 E_y = \frac{1}{15} \Rightarrow E_y = \frac{1}{15\chi_e\varepsilon_0}$$

$$\chi_e \varepsilon_0 E_z = \frac{1}{6} \Rightarrow E_z = \frac{1}{6\chi_e\varepsilon_0}$$

since $E_x + E_y + E_z = 10$

There fore

$$\left(\frac{1}{10} + \frac{1}{15} + \frac{1}{6}\right)\frac{1}{\chi_e\varepsilon_0} = 10$$

$$\Rightarrow \frac{1}{3\chi_e\varepsilon_0} = 10 \Rightarrow \boxed{\chi_e = \frac{1}{30\varepsilon_0}}$$

(b)

$\Rightarrow \bar{E} = E_x a_x + E_y a_y + E_z a_z$

$$E_x = \frac{1}{10\chi_e\varepsilon_0} = \frac{30\varepsilon_0}{10\varepsilon_0} = 3\ \text{V/m}$$

$$E_y = \frac{1}{15\chi_e\varepsilon_0} = \frac{30\varepsilon_0}{15\varepsilon_0} = 2\ \text{V/m}$$

Similarly

$$E_z = \frac{1}{6\chi_e\varepsilon_0} = \frac{30\varepsilon_0}{6\varepsilon_0} = 5\ \text{V/m}$$

Finally, $\bar{E} = 3a_x + 2a_y + 5a_z$

(c) $\bar{D} = \varepsilon_r\varepsilon_0\bar{E} = \varepsilon_0(1+\chi_e)\bar{E}$.

4. (10 points) In an oil chamber ($\varepsilon_r = 3.0$), a uniform electric field ($\mathbf{E}_1$) of strength 1000 V/m makes an angle $\theta_1 = \pi/6$ with the normal axis as depicted in **Fig. 2**. Next to which, there is a thick glass ($\varepsilon_r = 9.0$) mounted for shielding purpose.

Calculate the magnitude and direction of the electric field in the glass.

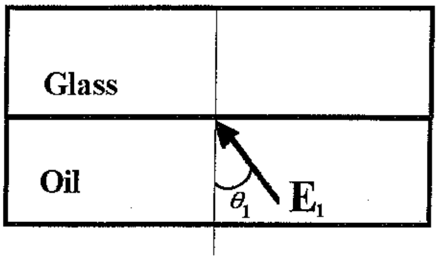

Fig. 2

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$\varepsilon_{r_1} = 3.0$ and $\varepsilon_{r_2} = 9.0$

$E_1 = 1000\,V/m$ @ $\theta_1 = \frac{\pi}{6}$ rad $= 30^\circ$

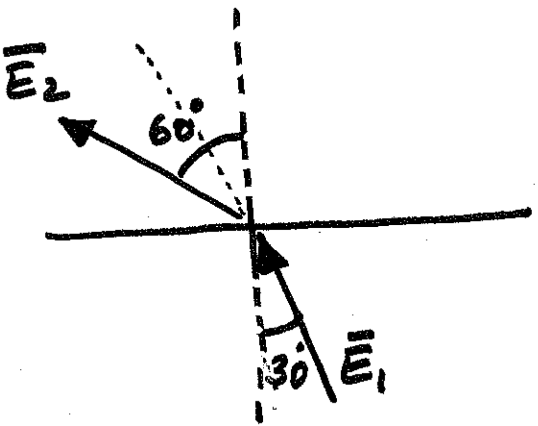
$E_{1t} = 1000 \sin 30^\circ = 500\,V/m$

$E_{1n} = 1000 \cos 30^\circ = 866.0254\,V/m$

$$\theta_2 = \tan^{-1}\left(\tan\theta_1 \frac{\varepsilon_{r2}}{\varepsilon_{r1}}\right) = \frac{\pi}{3} = 60^\circ$$

Since $E_{2t} = E_{1t} = 500\,V/m$

therefore
$$E_2 = \frac{E_{2t}}{\sin\theta_2} = \frac{500}{\sin 60^\circ} = 577.35\,V/m.$$



5. (10 points) The excess charge in a certain medium decreases to one-third of its initial value in 20 μs. (a) If the conductivity of the medium is 10^{-3} S/m, what is the dielectric constant of the medium? (b) What is the relaxation time? (c) After 40μs, what fraction of the charge will remain?

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(b) $@ t = t_1 = 20\mu s$, $\rho_v(t_1) = \frac{1}{3}\rho(0)$

since $\rho_v(t) = \rho(0)\exp(-t/T_r)$

therefore

$$\frac{\cancel{\rho(0)}}{3} = \cancel{\rho(0)}\exp(-t_1/T_r)$$

$$\Rightarrow T_r = \frac{t_1}{\ln(3)} = 18.205\mu s.$$

(a) $\sigma = 10^{-3}$ S/m

$$T_r = \varepsilon/\sigma = \varepsilon_0\varepsilon_r/\sigma$$

$$\varepsilon_r = \sigma T_r/\varepsilon_0 = 2058.9$$

(c) $@ t = t_2 = 40\mu s$, find $\rho(t_2)$

$$\rho(t_2) = \rho(0)\exp(-t_2/T_r)$$

$$= \rho(0)\exp(-40/18.205)$$

$$= (0.1111\cdots)\rho(0)$$

$$= \frac{1}{9}\rho(0) \text{ or } 11.11\% \text{ of } \rho(0).$$

6. (10 points) Let $\mathbf{J} = 400\sin\theta/(r^2+4)\mathbf{a}_r$ A/m^2. Find the total current flowing through that portion of the spherical surface $r = 1$, bounded by $0.1\pi < \theta < 0.3\pi, 0 < \phi < 2\pi$.

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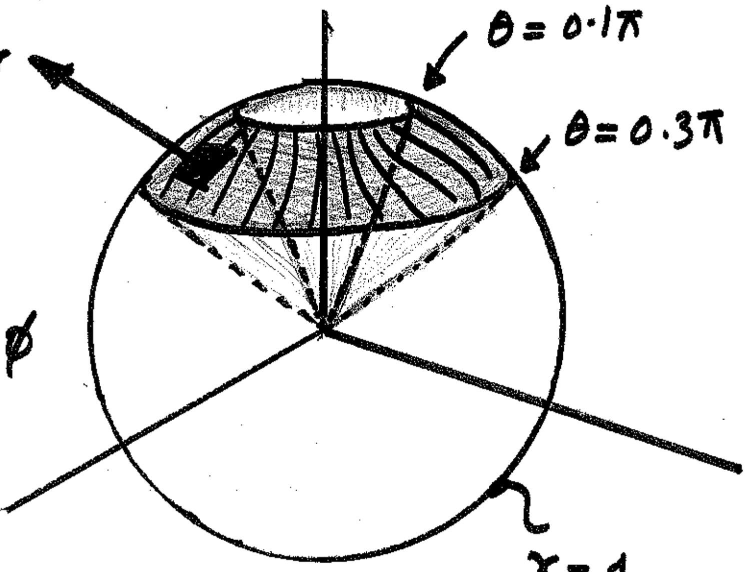
$$d\bar{S} = r^2 \sin\theta \, d\theta \, d\phi \, \mathbf{a}_r$$



$$I = \int \bar{J} \cdot d\bar{S}$$

$$= \int_{(r=1)} \frac{400\sin\theta}{r^2+4} r^2 \sin\theta \, d\theta \, d\phi$$

$$= \frac{400}{5} \int_{\phi=0}^{2\pi} \int_{\theta=0.1\pi}^{0.3\pi} \sin^2\theta \, d\theta \, d\phi \qquad 0 < \phi < 2\pi$$

$$= \frac{400}{5} \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0.1\pi}^{0.3\pi} \sin^2\theta \, d\theta$$

$$= 80\,(2\pi) \int_{\theta_1}^{\theta_2} \left(\frac{1-\cos 2\theta}{2} \right) d\theta$$

$$= 160\pi \left(\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right) \Bigg|_{\theta=0.1\pi}^{0.3\pi}$$

$$I = 112.26 \text{ A.}$$

7. (10 points) If volume charge density is given as $\rho_v = (\cos\omega t)/r^2$ C/m^3 in spherical coordinates, find **J**. It is reasonable to assume that **J** is not a function of θ or ϕ.

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$$\nabla \cdot \bar{J} = \frac{-\partial \rho_v(t)}{\partial t} \quad \text{———} \quad (1)$$

If $\bar{J}$ is not the function of θ and ϕ, then

$$\nabla \cdot \bar{J} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 J_r) \quad \text{———} \quad (2)$$

Note that $\dfrac{\partial \rho_v(t)}{\partial t} = \dfrac{-1}{r^2}\omega \sin\omega t$ ——— (3)

Combining (1), (2) and (3), we obtain.

$$\frac{\partial}{\partial r}(r^2 J_r) = \omega \sin \omega t$$

Integrating both sides w.r.t. "r", we get

$$r^2 J_r = \omega \sin\omega t \, r + c$$

The constant of integration, c, is equal to zero. This is obtained when we substitute r = 0. Finally

$$\boxed{\bar{J} = J_r a_r = \frac{\omega \sin\omega t}{r} a_r}$$

8. (10 points) Given the potential field

$$V = \frac{100\,x\,z}{x^2+4}$$

in free space. Find $\mathbf{D}$ at the surface $z = 0$.

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$$\bar{D} = \varepsilon_0 \bar{E}$$

$$= -\varepsilon_0 \nabla V$$

$$\nabla V = 100z \frac{\partial}{\partial x}\left(\frac{x}{x^2+4}\right) a_x + 0 \cdot a_y + \frac{100x}{x^2+4} a_z$$

@ $z=0$, we have

$$\nabla V = \frac{100x}{x^2+4} a_z$$

Therefore

$$\boxed{\bar{D} = -\frac{\varepsilon_0 100x}{x^2+4} a_z}$$

9. (10 points) Find the dielectric constant of a material in which the electric flux density is four times the polarization.

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$$\bar{D} = \varepsilon_0 \bar{E} + \bar{P}$$

Given $\bar{P} = \bar{D}/4$, we get

$$\bar{D} = \varepsilon_0 \bar{E} + \frac{\bar{D}}{4}$$

$$\Rightarrow \bar{D} = \frac{4}{3} \varepsilon_0 \bar{E}$$

$$\Rightarrow \boxed{\varepsilon_r = 4/3}$$

or $\varepsilon_r = 1 + \chi_e = 4/3$

$$\Rightarrow \boxed{\chi_e = \frac{1}{3}}$$

10. (10 points) A point charge Q is placed at the origin. Calculate the energy (W_E) stored in the region defined by $r > a$.

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We know that

$$W_E = \frac{1}{2}\varepsilon_0 \int |\bar{E}|^2 dv = \frac{1}{2}\int (\bar{D}\cdot\bar{E})dv$$

For the point charge, using Coulombs' Law,

$$\bar{E} = \frac{kQ}{r^2}a_r$$

$$\Rightarrow |\bar{E}|^2 = \frac{k^2Q^2}{r^4}$$

In spherical systems, $dv = r^2 \sin\theta \, d\theta \, d\phi \, dr$

$$W_E = \frac{1}{2}\varepsilon_0 \iiint \frac{k^2Q^2}{r^4}(r^2 \sin\theta \, d\theta \, d\phi \, dr)$$

$$= \frac{Q^2}{32\pi^2\varepsilon_0}\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi}\int_{r=a}^{\infty}\frac{1}{r^2}\sin\theta \, d\theta \, d\phi \, dr$$

$$\boxed{W_E = \frac{Q^2}{8\pi\varepsilon_0 a}}$$